PRACTICAL 5

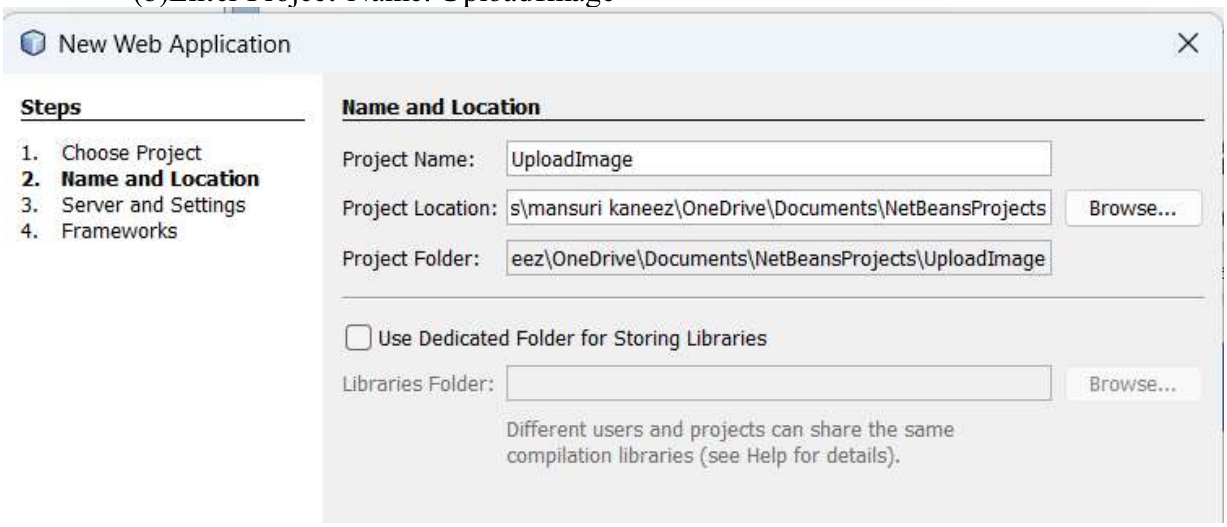
AIM: Develop application to download image/video from server or upload image/video

Part A: Create Web Service (Server Side)

STEP 1: Create New Project

(a).Click on File → New Project → Java Web → Web Application.

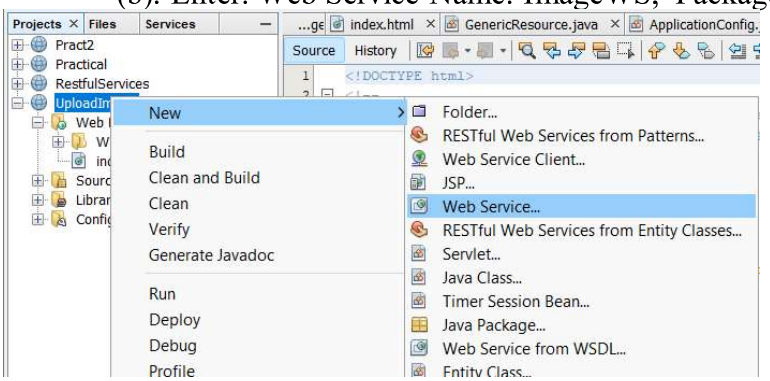
(b)Enter Project Name: UploadImage

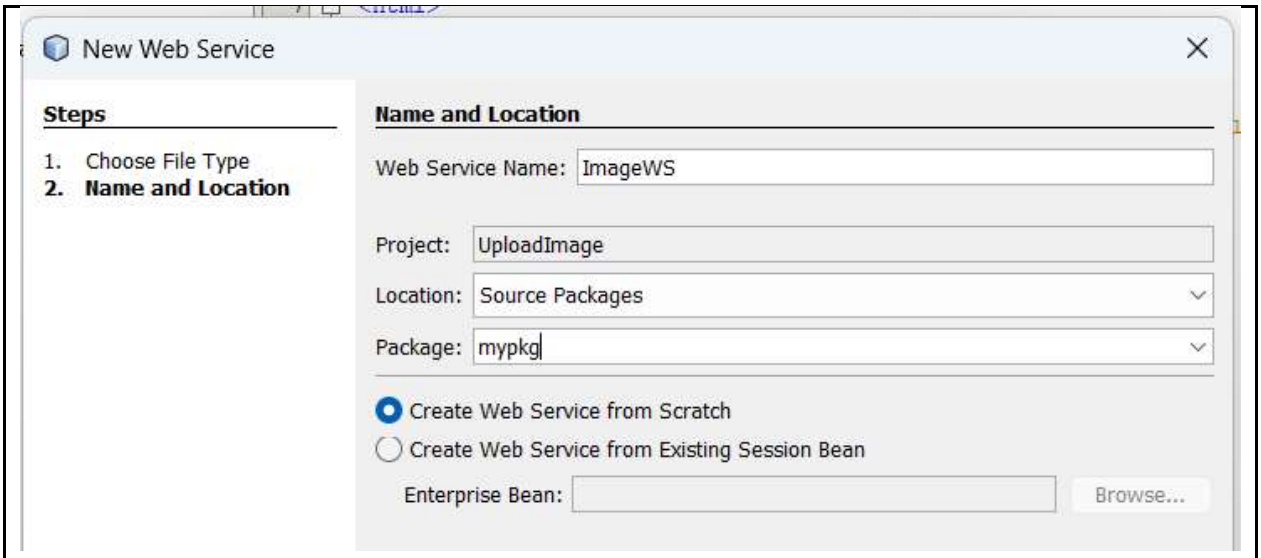


STEP 2: Create new Web service

(a). Right click on Project → New → Web Services → RESTful Web Services.

(b). Enter: Web Service Name: ImageWS, Package Name: mypkg





STEP 3: Write the following code in the web service class to accept image/video file:

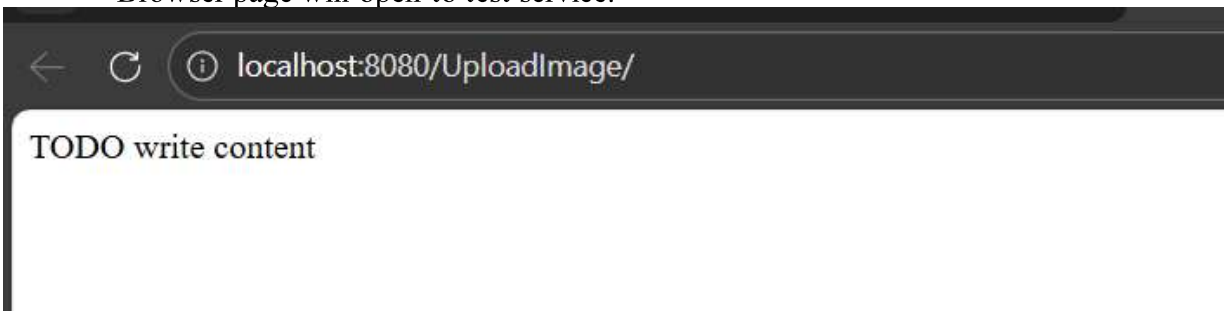
```
package mypkg;

import java.io.*;
import javax.jws.Oneway;
import javax.jws.WebService;
import javax.jws.WebMethod;
import javax.jws.WebParam;
import javax.xml.ws.soap.MTOM;

@MTOM(enabled = true, threshold = 60000)
@WebService(serviceName = "ImageWS")
public class ImageWS {

    @WebMethod(operationName = "upload")
    @Oneway
    public void upload(
            @WebParam(name = "Filename") String Filename,
            @WebParam(name = "ImageBytes") byte[] ImageBytes) {

        String filePath = "C:/MSC/upload/" + Filename;

        try {
            FileOutputStream fos = new FileOutputStream(filePath);
            BufferedOutputStream bos = new BufferedOutputStream(fos);
            bos.write(ImageBytes);
            bos.close();
            System.out.println("Received file: " + filePath);
        } catch (Exception ex) {
            System.out.println(ex);
        }
    }
```

```
    @WebMethod(operationName = "download")
    public byte[] download(
            @WebParam(name = "Filename") String Filename) {

        String filePath = "C:/MSC/upload/" + Filename;
        System.out.println("Sending file: " + filePath);

        try {
            File file = new File(filePath);
            FileInputStream fis = new FileInputStream(file);
            BufferedInputStream bis = new BufferedInputStream(fis);

            byte[] fileBytes = new byte[(int) file.length()];
            bis.read(fileBytes);
            bis.close();

            return fileBytes;
        } catch (Exception ex) {
            System.out.println(ex);
        }

        return null;
    }
}
```

STEP 4: Right click on Project → Clean and Build.

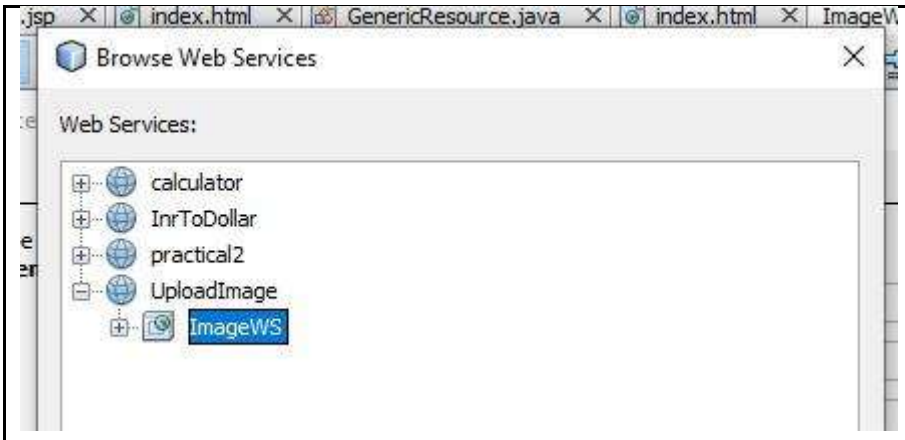
STEP 5: Right click on RESTful Service → Test RESTful Web Service.

Browser page will open to test service.

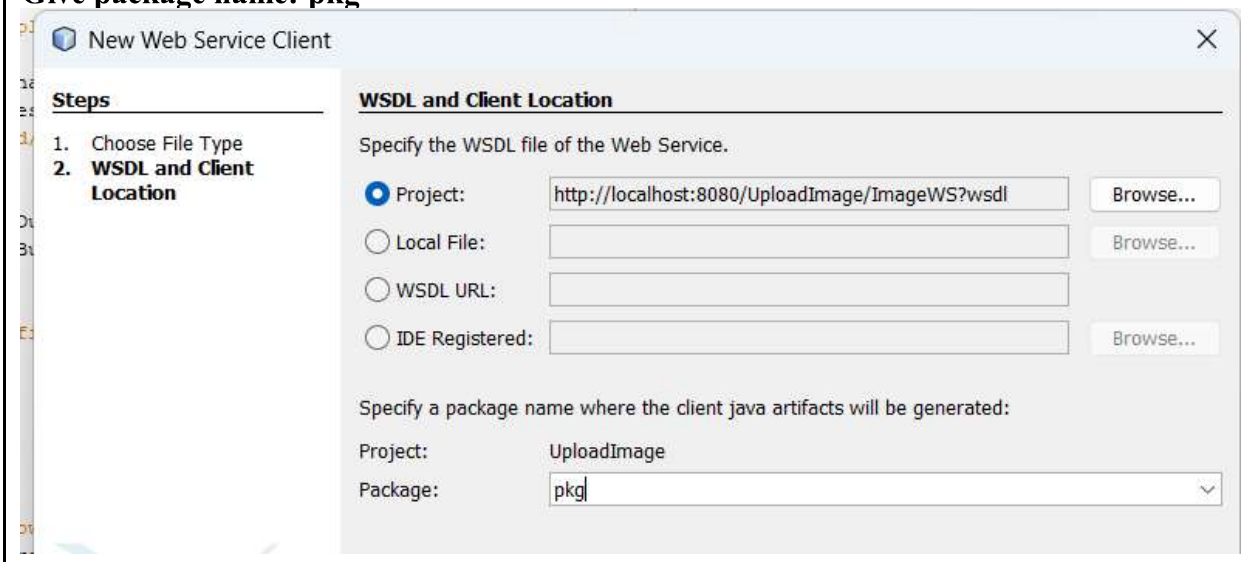


STEP 6: Create Web Service Client

(a).Browse ImageWS from the Project

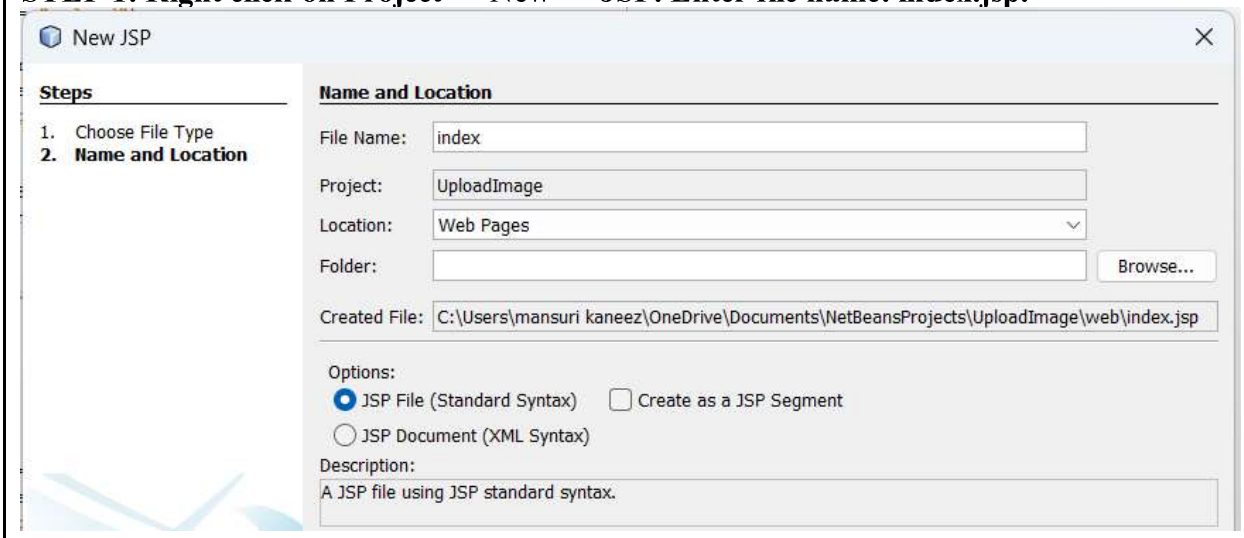


Give package name: pkg



Part B: Create Client (JSP Page)

STEP 1: Right click on Project → New → JSP. Enter file name: index.jsp.



STEP 2: Write the following code in the file :

```
<%@page import="java.io.BufferedOutputStream"%>
```

```
<%@page import="java.io.FileOutputStream"%>
<%@page import="java.io.FileInputStream"%>
<%@page import="java.io.BufferedInputStream"%>
<%@page import="java.io.File"%>
<%@page import="javax.xml.ws.soap.MTOMFeature"%>
<%@page contentType="text/html" pageEncoding="UTF-8"%>

<!DOCTYPE html>
<html>
<head>
    <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
    <title>Image Upload & Download</title>
</head>

<body>

    <h2>Upload Image to Server</h2>
    <hr/>

    <%-- Upload Section --%>
    <%
        try {
            pkg.ImageWS_Service service = new pkg.ImageWS_Service();
            pkg.ImageWS port = service.getImageWSPort(new MTOMFeature(60000));

            String filePath = "C:/Picture/flower.jpg";
            File file = new File(filePath);

            FileInputStream fis = new FileInputStream(file);
            BufferedInputStream bis = new BufferedInputStream(fis);

            String filename = file.getName();
            byte[] imageBytes = new byte[(int) file.length()];
            bis.read(imageBytes);

            port.upload(filename, imageBytes);
            bis.close();

            out.println("<b>File Uploaded Successfully:</b> " + filePath);

        } catch (Exception ex) {
            ex.printStackTrace();
        }
    %>

    <hr/>
    <h2>Download Image from Server</h2>
    <hr/>

    <%-- Download Section --%>
    <%
        try {
            pkg.ImageWS_Service service = new pkg.ImageWS_Service();
            pkg.ImageWS port = service.getImageWSPort();
```

```
            String filename = "flower.jpg";
            String filePath = "C:/Picture/download/" + filename;

            byte[] fileBytes = port.download(filename);

            FileOutputStream fos = new FileOutputStream(filePath);
            BufferedOutputStream bos = new BufferedOutputStream(fos);

            bos.write(fileBytes);
            bos.close();

            out.println("<b>File Downloaded Successfully:</b> " + filePath);

        } catch (Exception ex) {
            ex.printStackTrace();
        }
    %>

    <hr/>

</body>
</html>
```

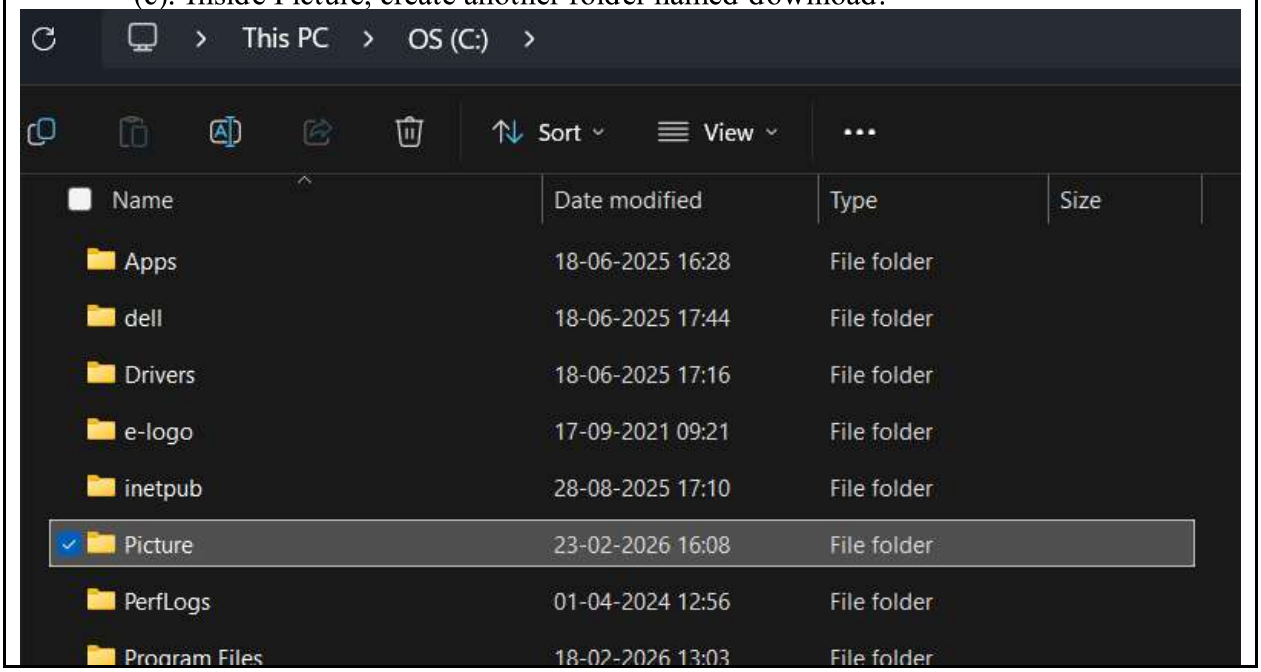
PART C: Create Folder in System

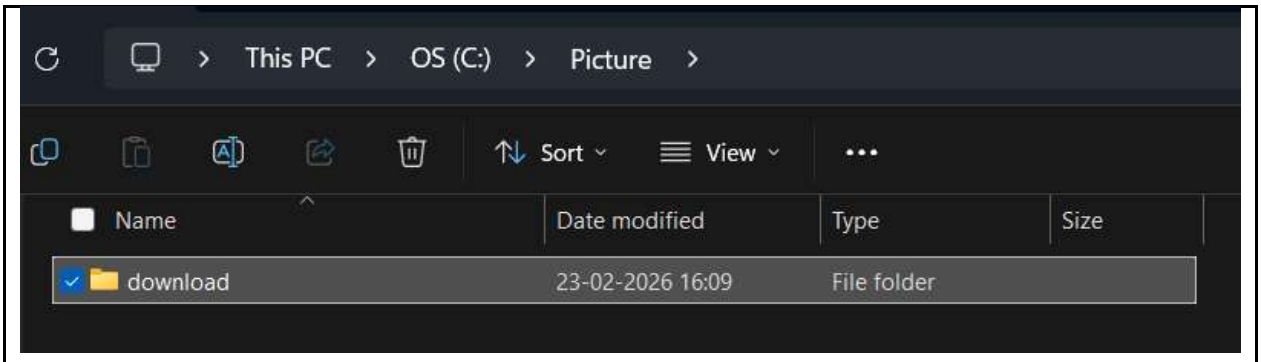
STEP 1: Create Folders

(a). Go to This PC → C Drive.

(b).Create new folder named Picture.

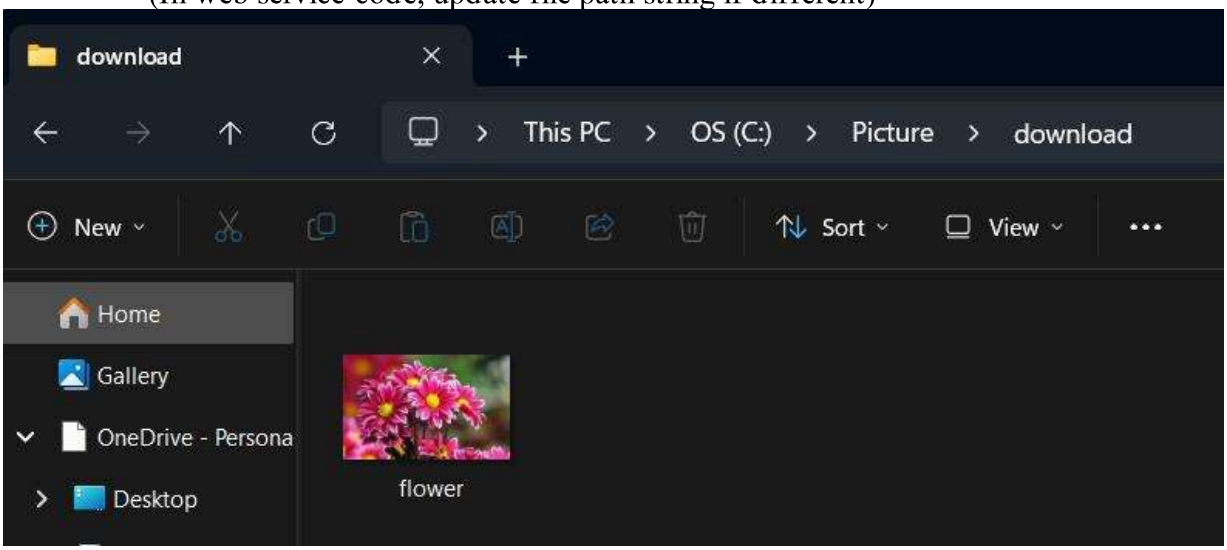
(c). Inside Picture, create another folder named download.





STEP 2: Image download

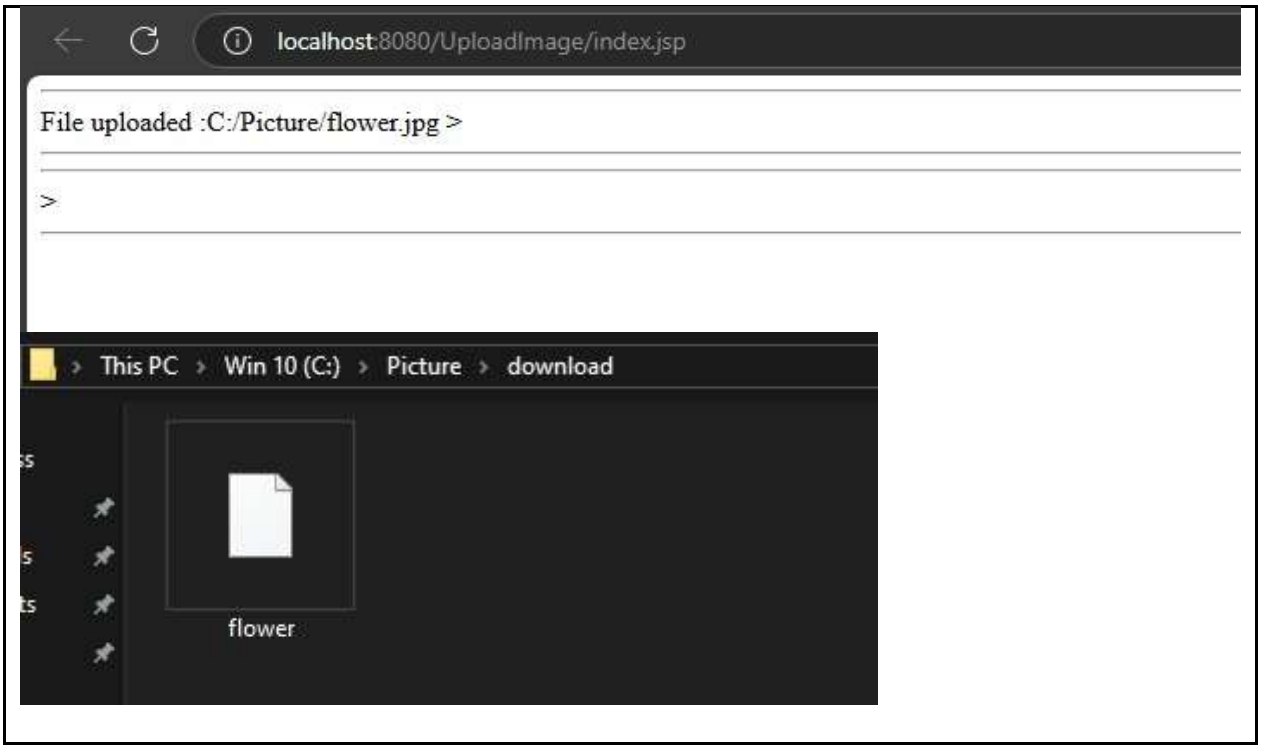
(a). Download any sample image (example: flower image) from browser.
(b). Save and rename it as flower.jpg inside C:\Picture\download.
(In web service code, update file path string if different)



STEP 3: Deploy the Project

STEP 4: Run index.jsp file.

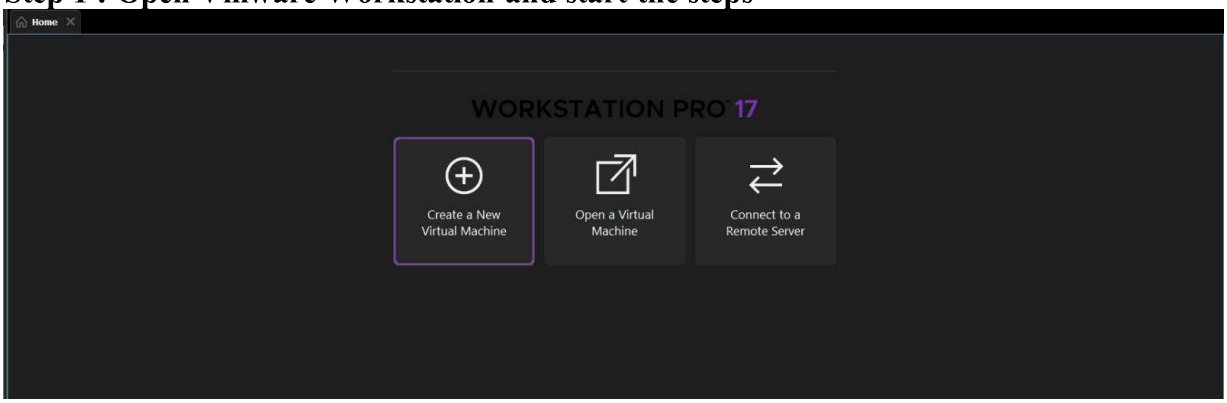
Verify that file is stored in C:\Picture\download.


localhost:8080/UploadImage/index.jsp
File uploaded :C:/Picture/flower.jpg >
>
This PC > Win 10 (C:) > Picture > download
flower

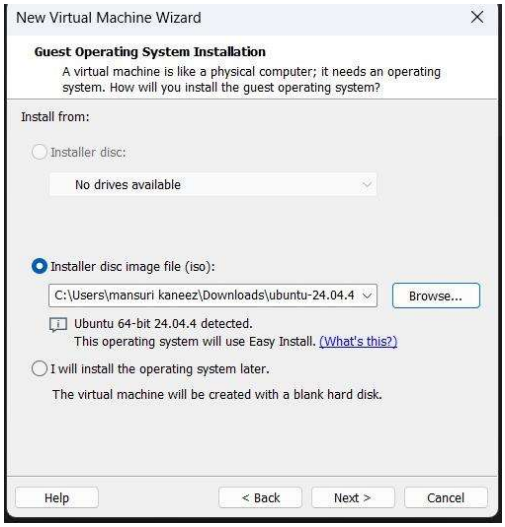
PRACTICAL 6

AIM: Installation and Configuration of virtualization using KVM

Step 1 : Open Vmware Workstation and start the steps

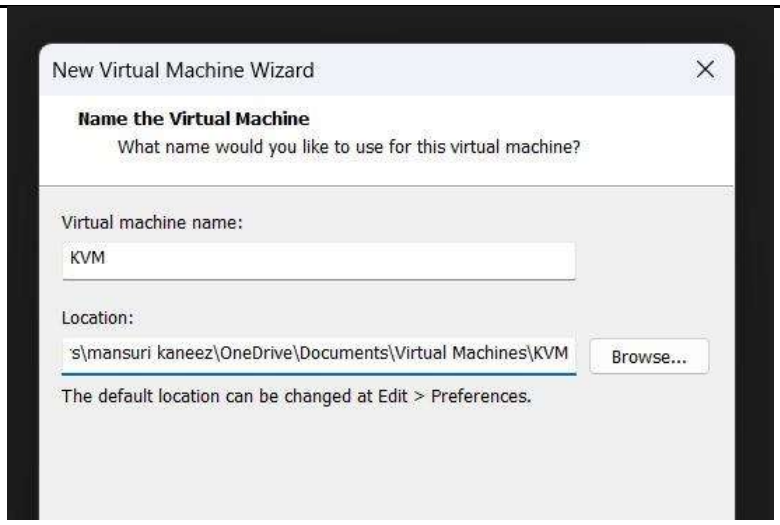


Step 2 : Click on new Virtual Machine

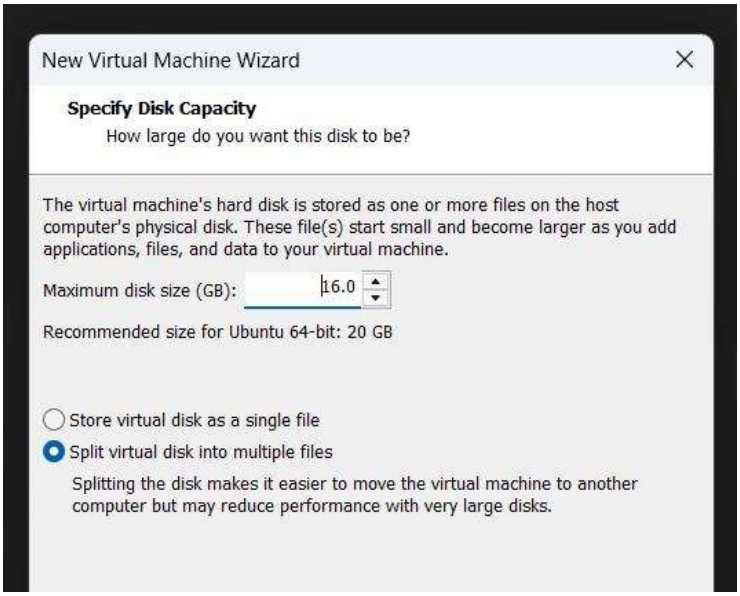


Step 3: Once the folder has been selected click on next, fill up the necessary details

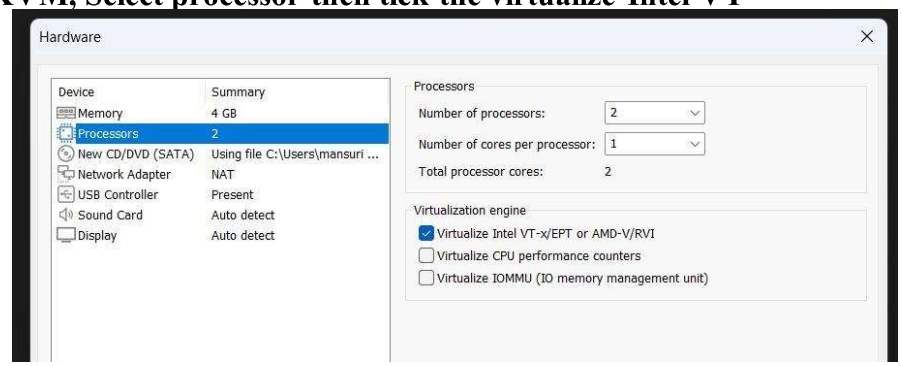
Step 4: Click on next and give a name you your virtual machine



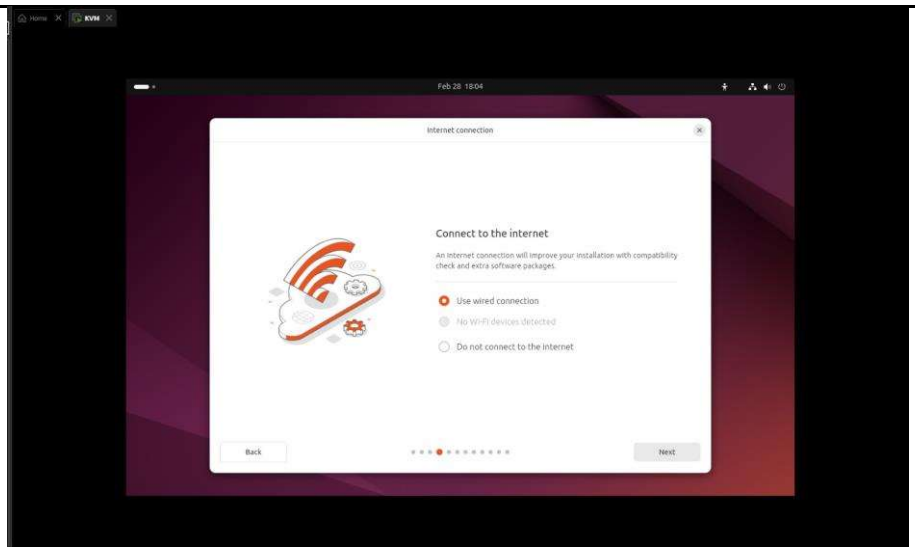
Step 5: Click on next and give disk size as 16 gb keep the spilt virtual disk into multiple file option selected and click on next



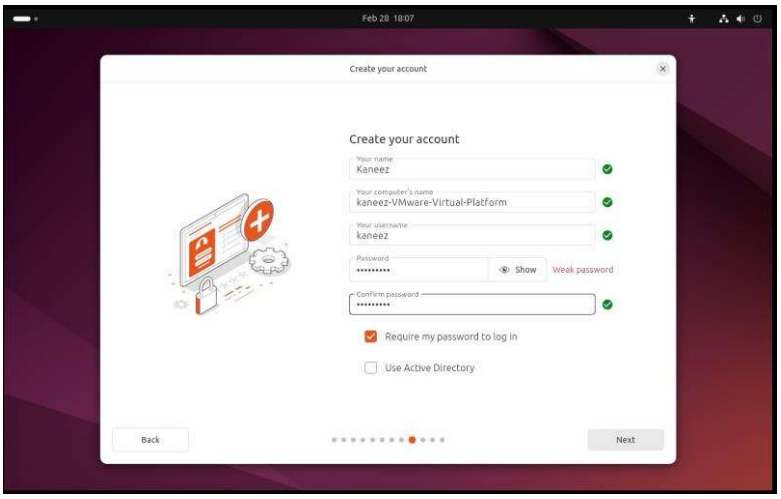
Step 6: Click on Customize Hardware
To enable KVM, Select processor then tick the virtualize Intel VT



Step 7: Click on Finish to start your Ubuntu

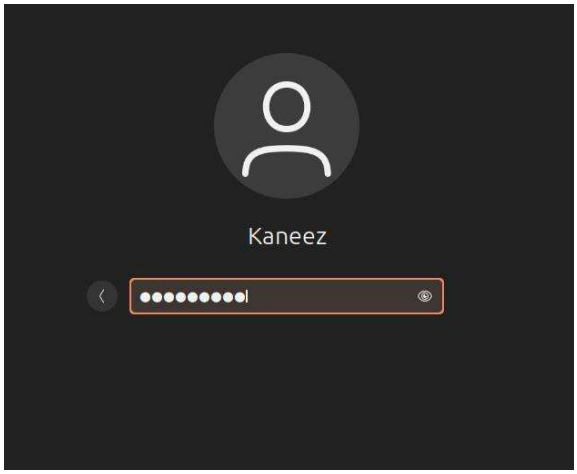


Keep clicking on Next until it asks for details:



Again click next, then restart

Step 8 : Login to your Ubuntu



Step 9: now right click and select open terminal

```
Home  KVM
Feb 28 23:53
kaneez@kaneez-VMware-Virtual-Platform: ~/Desktop
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

kaneez@kaneez-VMware-Virtual-Platform:~/Desktop$
```

Now enter this command to update your Ubuntu
$ sudo apt update:

```
kaneez@kaneez-VMware-Virtual-Platform:~/Desktop$ sudo apt update
[sudo] password for kaneez:
Hit:1 http://in.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
13 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

$ sudo apt upgrade –y

```
kaneez@kaneez-VMware-Virtual-Platform:~/Desktop$ sudo apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
Get more security updates through Ubuntu Pro with 'esm-apps' enabled:
  libzvbi-common libcjson1 libpostproc57 libavcodec60
  libgstreamer-plugins-bad1.0-0 libzvbi0t64 libavutil58 libswscale7
  libswresample4 libavformat60 libavfilter9
Learn more about Ubuntu Pro at https://ubuntu.com/pro
The following packages will be upgraded:
  cloud-init libfprint-2-2 libfprint-2-tod1 libgphoto2-6t64 libgphoto2-l10n libgphoto2-port12t64 libmtp-common libmtp-runtime libmtp9t64
  python3-software-properties software-properties-common software-properties-gtk systemd-hwe-hwdb
13 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
Need to get 2,084 kB of archives.
After this operation, 20.5 kB of additional disk space will be used.
Get:1 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 systemd-hwe-hwdb all 255.1.7 [3,716 B]
Get:2 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 libfprint-2-tod1 amd64 1:1.94.7+tod1-0ubuntu5~24.04.5 [64.5 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 libfprint-2-2 amd64 1:1.94.7+tod1-0ubuntu5~24.04.5 [264 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 libgphoto2-port12t64 amd64 2.5.31-2.1ubuntu1 [58.8 kB]
Get:5 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 libgphoto2-6t64 amd64 2.5.31-2.1ubuntu1 [735 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 libgphoto2-l10n all 2.5.31-2.1ubuntu1 [15.1 kB]
Get:7 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 libmtp-common all 1.1.21-3.1ubuntu1 [19.7 kB]
Get:8 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 libmtp-runtime amd64 1.1.21-3.1ubuntu1 [6,046 B]
Get:9 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 libmtp9t64 amd64 1.1.21-3.1ubuntu1 [162 kB]
Get:10 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 software-properties-common all 0.99.49.4 [14.4 kB]
Get:11 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 software-properties-gtk all 0.99.49.4 [83.2 kB]
Get:12 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 python3-software-properties all 0.99.49.4 [30.0 kB]
```

After the upgrade it will ask you to restart your ubuntu, click on restart and login in your account once again and restart your terminal.

Type the following commands one by one:

$ sudo apt install cpu-checker

```
kaneez@kaneez-VMware-Virtual-Platform:~/Desktop$ sudo apt install cpu-checker
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  msr-tools
The following NEW packages will be installed:
  cpu-checker msr-tools
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
Need to get 15.8 kB of archives.
After this operation, 65.5 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://in.archive.ubuntu.com/ubuntu noble/main amd64 msr-tools amd64 1.3-5build1 [9,610 B]
Get:2 http://in.archive.ubuntu.com/ubuntu noble/main amd64 cpu-checker amd64 0.7-1.3build2 [6,148 B]
Fetched 15.8 kB in 2s (8,646 B/s)
Selecting previously unselected package msr-tools.
(Reading database ... 152183 files and directories currently installed.)
Preparing to unpack .../msr-tools_1.3-5build1_amd64.deb ...
Unpacking msr-tools (1.3-5build1) ...
Selecting previously unselected package cpu-checker.
Preparing to unpack .../cpu-checker_0.7-1.3build2_amd64.deb ...
Unpacking cpu-checker (0.7-1.3build2) ...
Setting up msr-tools (1.3-5build1) ...
Setting up cpu-checker (0.7-1.3build2) ...
Processing triggers for man-db (2.12.0-4build2) ...
kaneez@kaneez-VMware-Virtual-Platform:~/Desktop$
```

$ sudo apt install qemu-kvm virt-manager libvirt-daemon-system libvirt-clients bridge-utils- y

```
kaneez@kaneez-VMware-Virtual-Platform: ~/Desktop
kaneez@kaneez-VMware-Virtual-Platform:~/Desktop$ sudo apt install qemu-kvm virt-manager libvirt-daemon-system libvirt-clients bridge-utils -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Note, selecting 'qemu-system-x86' instead of 'qemu-kvm'
The following additional packages will be installed:
  dmeventd gir1.2-ayatanaappindicator3-0.1 gir1.2-gtk-vnc-2.0 gir1.2-gtksource-4 gir1.2-libosinfo-1.0 gir1.2-libvirt-glib-1.0 gir1.2-spiceclientglib-2.0
  gir1.2-spiceclientgtk-3.0 ipxe-qemu ipxe-qemu-256k-compat-efi-roms libaio1t64 libboost-iostreams1.83.0 libboost-thread1.83.0 libburn4t64 libcacard0 libdaxctl1
  libdevmapper-event1.02.1 libfdt1 libgtk-vnc-2.0-0 libgtksourceview-4-0 libgtksourceview-4-common libgvnc-1.0-0 libiscsi7 libisoburn1t64 libisofs6t64
  libjack-jackd2-0 liblvm2cmd2.03 libndctl6 libnss-mymachines libosinfo-1.0-0 libosinfo-l10n libphodav-3.0-0 libphodav-3.0-common libpmem1 libpmemobj1 librados2
  librbd1 librdmacm1t64 libsdl2-2.0-0 libslirp0 libspice-client-glib-2.0-8 libspice-client-gtk-3.0-5 libspice-server1 libtpms0 liburing2 libusbredirhost1t64
  libusbredirparser1t64 libvirglrenderer1 libvirt-daemon libvirt-daemon-config-network libvirt-daemon-config-nwfilter libvirt-daemon-driver-qemu
  libvirt-daemon-system-systemd libvirt-glib-1.0-0 libvirt-glib-1.0-data libvirt-l10n libvirt0 libxml2-utils libyajl2 lvm2 mdevctl osinfo-db ovmf
  python3-gi-cairo python3-libvirt python3-libxml2 qemu-block-extra qemu-system-common qemu-system-data qemu-system-gui qemu-system-modules-opengl
  qemu-system-modules-spice qemu-utils seabios spice-client-glib-usb-acl-helper swtpm swtpm-tools systemd-container thin-provisioning-tools virt-viewer virtinst
  xorriso
Suggested packages:
  ifupdown jackd2 gstreamer1.0-plugins-bad libvirt-clients-qemu libvirt-login-shell libvirt-daemon-driver-storage-gluster
  libvirt-daemon-driver-storage-iscsi-direct libvirt-daemon-driver-storage-rbd libvirt-daemon-driver-storage-zfs libvirt-daemon-driver-lxc
  libvirt-daemon-driver-vbox libvirt-daemon-driver-xen numad passt auditd nfs-common open-iscsi pm-utils systemtap zfsutils samba vde2 trousers python3-guestfs
  ssh-askpass python3-argcomplete xorriso-tcltk jigit cdck
The following NEW packages will be installed:
  bridge-utils dmeventd gir1.2-ayatanaappindicator3-0.1 gir1.2-gtk-vnc-2.0 gir1.2-gtksource-4 gir1.2-libosinfo-1.0 gir1.2-libvirt-glib-1.0
  gir1.2-spiceclientglib-2.0 gir1.2-spiceclientgtk-3.0 ipxe-qemu ipxe-qemu-256k-compat-efi-roms libaio1t64 libboost-iostreams1.83.0 libboost-thread1.83.0
  libburn4t64 libcacard0 libdaxctl1 libdevmapper-event1.02.1 libfdt1 libgtk-vnc-2.0-0 libgtksourceview-4-0 libgtksourceview-4-common libgvnc-1.0-0 libiscsi7
  libisoburn1t64 libisofs6t64 libjack-jackd2-0 liblvm2cmd2.03 libndctl6 libnss-mymachines libosinfo-1.0-0 libosinfo-l10n libphodav-3.0-0 libphodav-3.0-common
  libpmem1 libpmemobj1 librados2 librbd1 librdmacm1t64 libsdl2-2.0-0 libslirp0 libspice-client-glib-2.0-8 libspice-client-gtk-3.0-5 libspice-server1 libtpms0
  liburing2 libusbredirhost1t64 libusbredirparser1t64 libvirglrenderer1 libvirt-clients libvirt-daemon libvirt-daemon-config-network
  libvirt-daemon-config-nwfilter libvirt-daemon-driver-qemu libvirt-daemon-system libvirt-daemon-system-systemd libvirt-glib-1.0-0 libvirt-glib-1.0-data
  libvirt-l10n libvirt0 libxml2-utils libyajl2 lvm2 mdevctl osinfo-db ovmf python3-gi-cairo python3-libvirt python3-libxml2 qemu-block-extra qemu-system-common
  qemu-system-data qemu-system-gui qemu-system-modules-opengl qemu-system-modules-spice qemu-system-x86 qemu-utils seabios spice-client-glib-usb-acl-helper swtpm
  swtpm-tools systemd-container thin-provisioning-tools virt-manager virt-viewer virtinst xorriso
0 upgraded, 87 newly installed, 0 to remove and 0 not upgraded.
Need to get 47.6 MB of archives.
After this operation, 212 MB of additional disk space will be used.
Get:1 http://in.archive.ubuntu.com/ubuntu noble/main amd64 bridge-utils amd64 1.7.1-1ubuntu2 [33.9 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 libdevmapper-event1.02.1 amd64 2:1.02.185-3ubuntu3.2 [12.6 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 libaio1t64 amd64 0.3.113-6build1.1 [7,210 B]
Get:4 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 liblvm2cmd2.03 amd64 2.03.16-3ubuntu3.2 [797 kB]
Get:5 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 dmeventd amd64 2:1.02.185-3ubuntu3.2 [38.0 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu noble/main amd64 gir1.2-ayatanaappindicator3-0.1 amd64 0.5.93-1build3 [5,838 B]
Get:7 http://in.archive.ubuntu.com/ubuntu noble/universe amd64 libgvnc-1.0-0 amd64 1.3.1-1build2 [67.1 kB]
```

$ sudo systemctl enable libvirtd

$ sudo systemctl start libvirtd

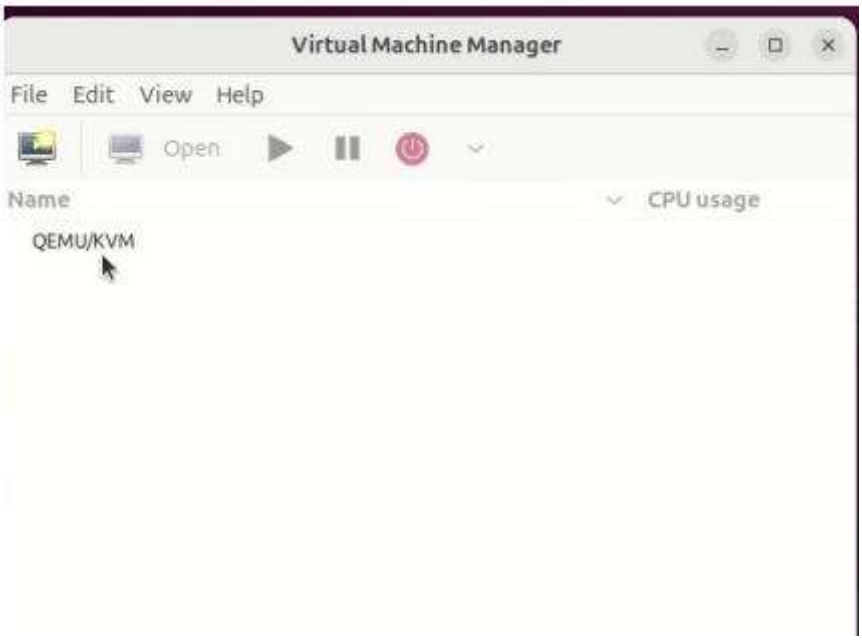
$ sudo usermod -aG kvm kaneez(your user name)

$ sudo usermod -aG libvirt kaneez (your user name)

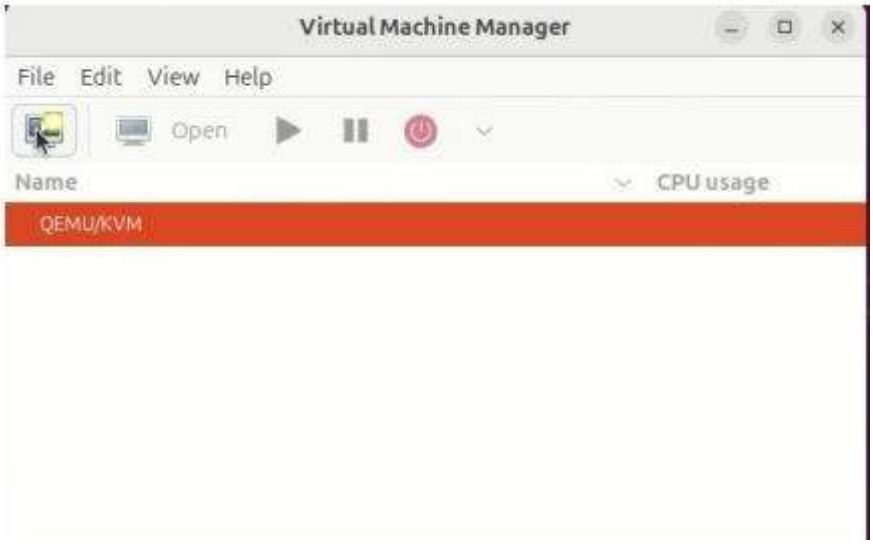
$ sudo systemctl status libvirtd

```
kaneez@kaneez-VMware-Virtual-Platform:~/Desktop$ sudo systemctl start libvirtd
kaneez@kaneez-VMware-Virtual-Platform:~/Desktop$ sudo systemctl enable libvirtd
kaneez@kaneez-VMware-Virtual-Platform:~/Desktop$ sudo systemctl start libvirtd
kaneez@kaneez-VMware-Virtual-Platform:~/Desktop$ sudo usermod -aG kvm kaneez
kaneez@kaneez-VMware-Virtual-Platform:~/Desktop$ sudo usermod -aG libvirt kaneez
kaneez@kaneez-VMware-Virtual-Platform:~/Desktop$ sudo systemctl status libvirtd
● libvirtd.service - libvirt legacy monolithic daemon
     Loaded: loaded (/usr/lib/systemd/system/libvirtd.service; enabled; preset: enabled)
     Active: active (running) since Sun 2026-03-01 00:06:41 IST; 1min 26s ago
TriggeredBy: ● libvirtd-ro.socket
             ● libvirtd-admin.socket
             ● libvirtd.socket
        Docs: man:libvirtd(8)
              https://libvirt.org/
    Main PID: 13877 (libvirtd)
       Tasks: 22 (limit: 32768)
      Memory: 12.8M (peak: 14.3M)
         CPU: 977ms
      CGroup: /system.slice/libvirtd.service
              ├─ 8273 /usr/sbin/dnsmasq --conf-file=/var/lib/libvirt/dnsmasq/default.conf --leasefile-ro --dhcp-script=/usr/lib/libvirt/libvirt_leaseshelper
              ├─ 8274 /usr/sbin/dnsmasq --conf-file=/var/lib/libvirt/dnsmasq/default.conf --leasefile-ro --dhcp-script=/usr/lib/libvirt/libvirt_leaseshelper
              └─13877 /usr/sbin/libvirtd --timeout 120

Mar 01 00:06:41 kaneez-VMware-Virtual-Platform systemd[1]: Starting libvirtd.service - libvirt legacy monolithic daemon...
Mar 01 00:06:41 kaneez-VMware-Virtual-Platform systemd[1]: Started libvirtd.service - libvirt legacy monolithic daemon.
Mar 01 00:06:41 kaneez-VMware-Virtual-Platform dnsmasq[8273]: read /etc/hosts - 8 names
Mar 01 00:06:41 kaneez-VMware-Virtual-Platform dnsmasq[8273]: read /var/lib/libvirt/dnsmasq/default.addnhosts - 0 names
Mar 01 00:06:41 kaneez-VMware-Virtual-Platform dnsmasq-dhcp[8273]: read /var/lib/libvirt/dnsmasq/default.hostsfile
Mar 01 00:06:41 kaneez-VMware-Virtual-Platform libvirtd[13877]: libvirt version: 10.0.0, package: 10.0.0-2ubuntu8.11 (Ubuntu)
```

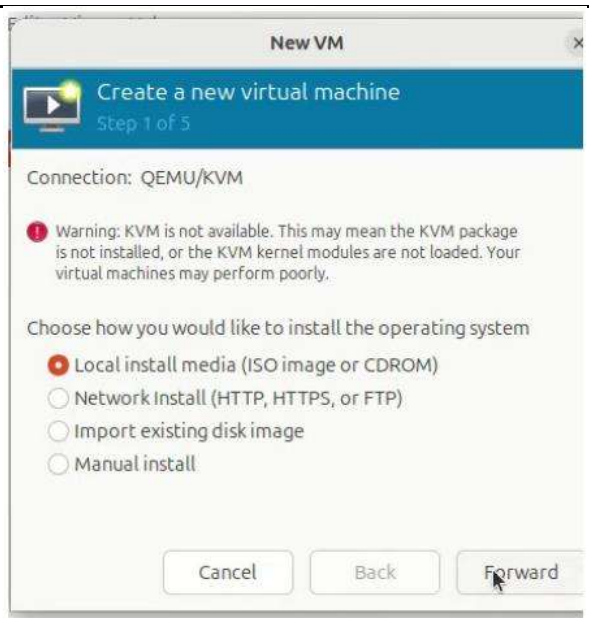
Now go to show apps and type vmm



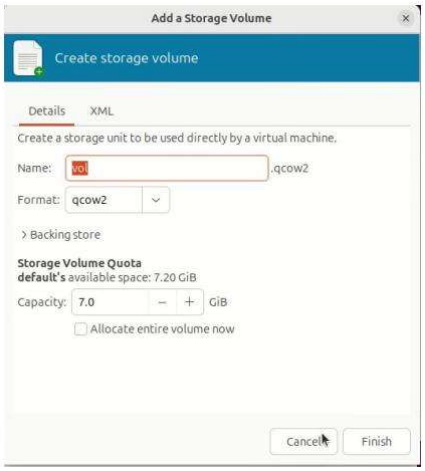
Click on the icon as shown



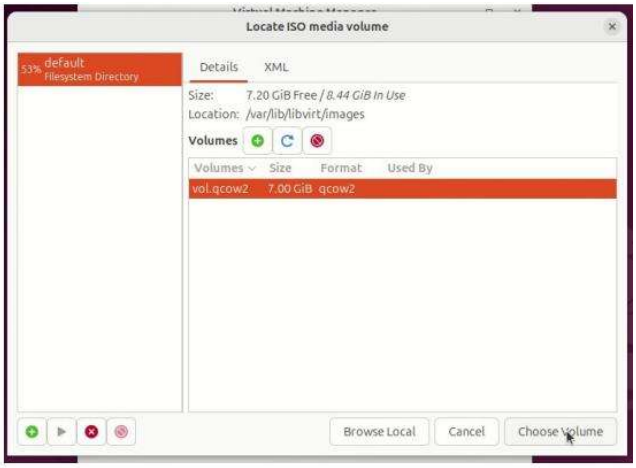
Now Click on Forward



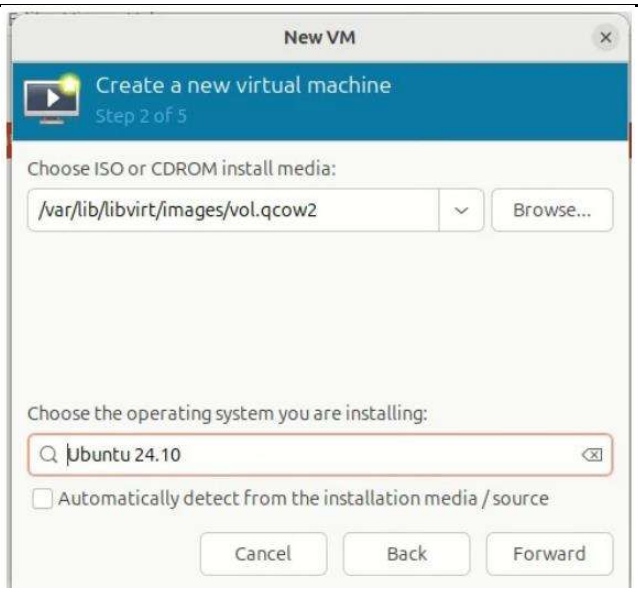
Click on Browse the click on the + sign
Click on Finish



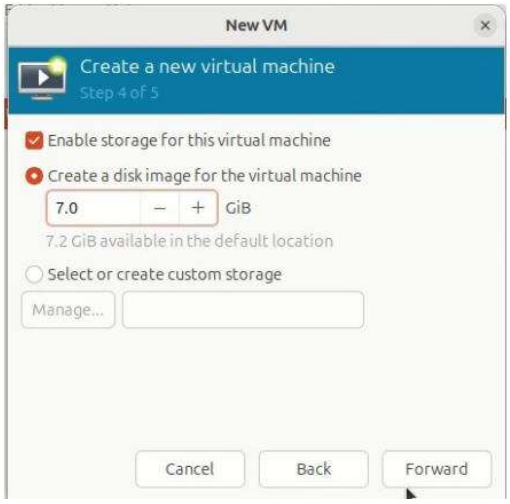
Now click on Choose volume



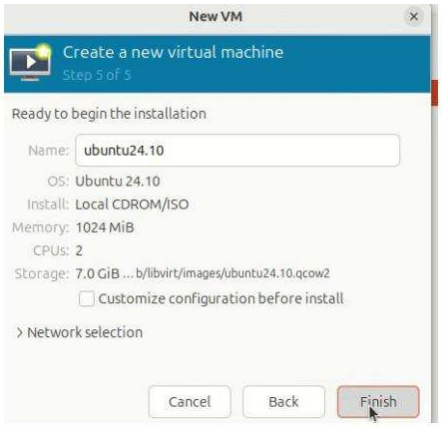
Now uncheck the box and search your installed ubuntu and then Click forward

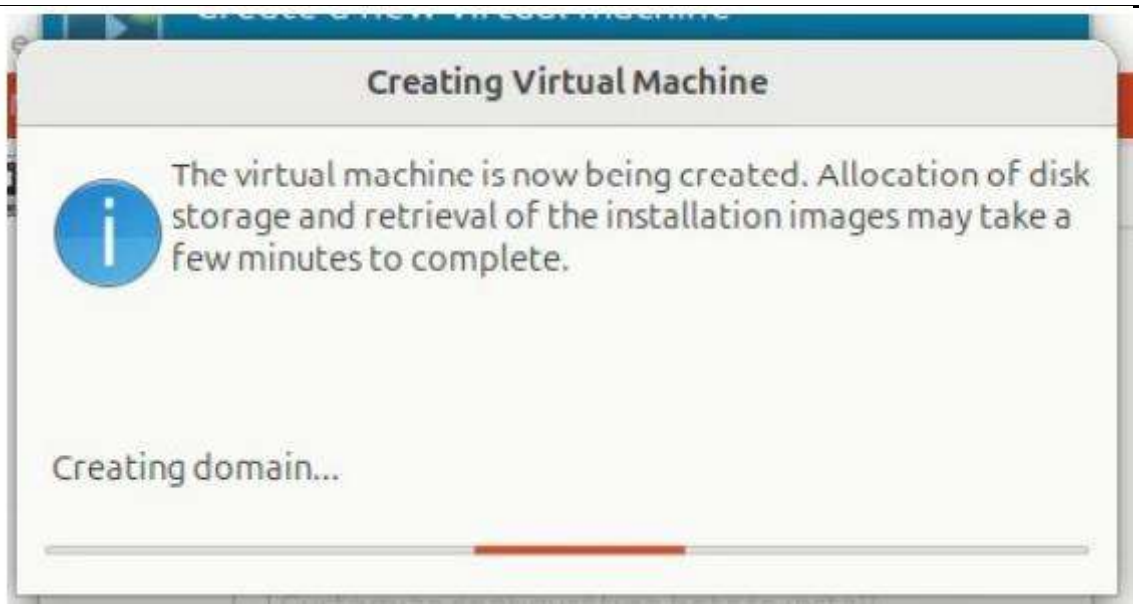
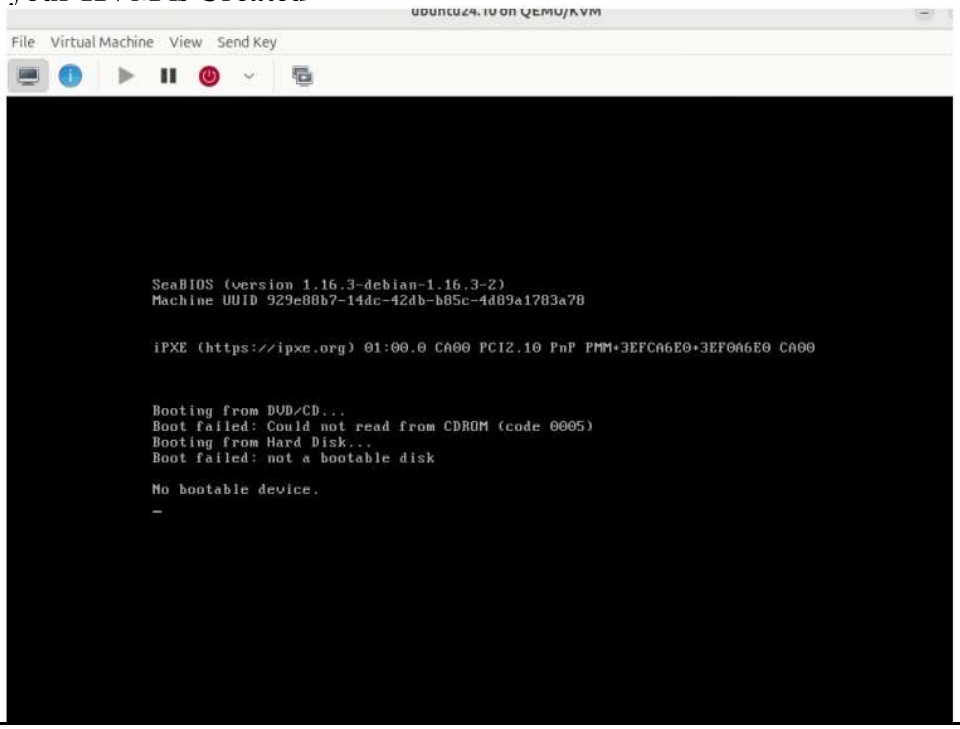


Now change your memory to 1024 and click forward



Click finish.





And then your KVM is Created